

Gravity Books Data Analysis and Optimization Strategies

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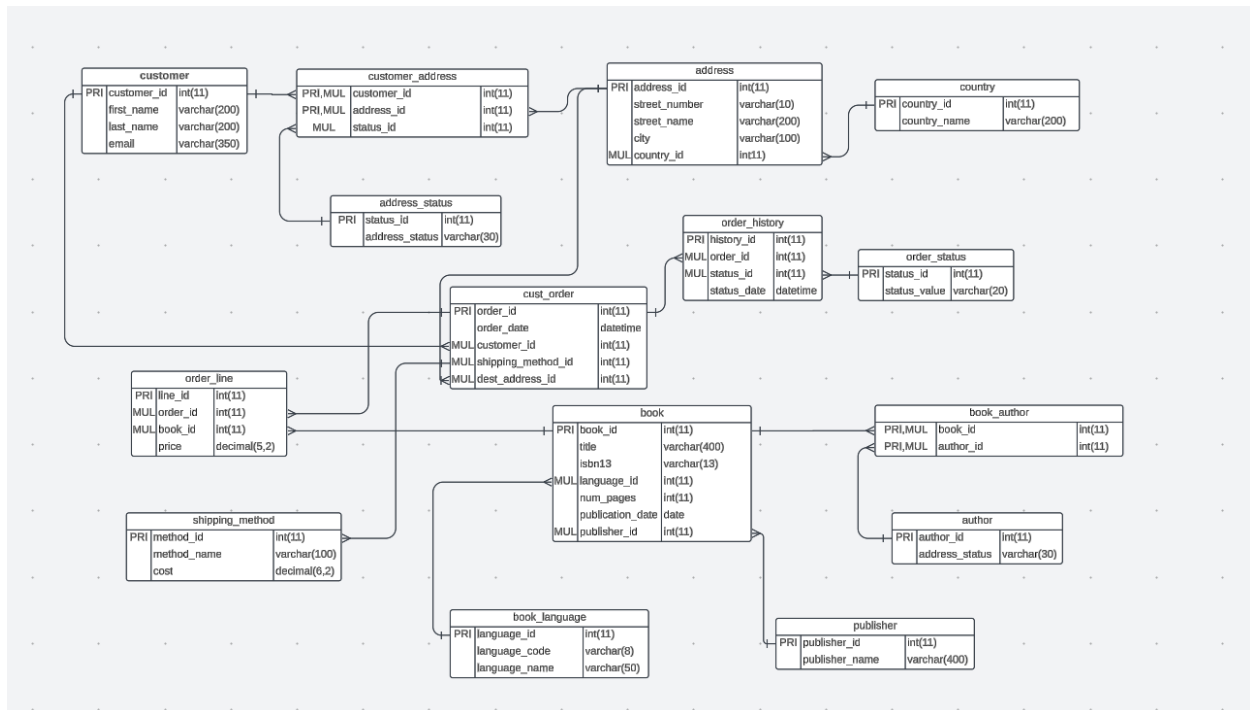
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Entity Relation Diagram:

The Entity Relation Diagram of the gravity books database is shown in the below diagram.

it has 15 tables and the relation between these tables is denoted by crow foot notation.

Two tables are connected by the common columns in those tables. PRI represents the primary key. MUL represents the foreign key.



1. Which language should they discontinue stocking?

Query used

SQL statement:

```
SELECT book.language_id,book_language.language_name,COUNT(order_line.book_id)
AS TotalOrders

FROM book

JOIN book_language

ON book.language_id = book_language.language_id

LEFT JOIN order_line ON book.book_id = order_line.book_id

GROUP BY book.language_id,book_language.language_name

ORDER BY TotalOrders ASC

LIMIT 10;
```

Explanation:

This query gives us the least amount of books sold for a particular language. I selected language_id, and language_name and counted the total number of orders for a language. I used the count aggregate to see how many orders were placed for a specific language. These

selected details are not present in a single table so I just joined multiple tables by using the common column in them. After that, I arranged them by language_id and language_name. I had them in ascending order to get the lower number of books sold first. After that, as there is no limitation I will get all the columns for all languages to solve that I use the limit option so that I can obtain the number of columns I want.

Why this query?

This query uses the demand for books with language as a different option, which gives us the least sold books. When we want to discontinue a language we should always look for the least sell books. When I used this query I got the number of books sold per language. So I think this is the best query to gather answers for the question.

Result table:

language_id	language_name	TotalOrders
20	Malaysian	0
21	Galician	0
25	Turkish	0
22	Welsh	0
24	Norwegian	1
17	Serbian	1
8	Middle English	1
26	Gaelic	1
19	Russian	1
12	Arabic	1

Explanation for output table:

The table shows us the language_id, language_name, and total_orders as the rows and we can see that we have written a query to get the orders per language so in this query we see the total orders per language. We have 10 languages as we have used a limit of 10 we can choose the language.

Recommendation:

I think they have to look out for the languages that they sell more and increase the production on that also it is good to discontinue in these places because, with the less income, we cannot run the warehouse.

2. What author should they pick for a special promotion?

Query used**SQL statement:**

```
SELECT author.author_name, COUNT(order_line.line_id) AS total_books  
  
FROM author  
  
JOIN book_author ON author.author_id = book_author.author_id  
  
JOIN order_line ON book_author.book_id = order_line.book_id  
  
GROUP BY author.author_name
```

```
ORDER BY total_books DESC
```

```
LIMIT 1;
```

Explanation:

In this query, we have selected the author's name and the total number of books sold by the author. As the details we not in a single table we combined three tables named author, book_author and orderline. We used book_author to connect the author table and order line because there is no common row in the table book_author and orderline. So we first joined author and book_author by using author_id and we joined book_author and order_line by using book_id. Later we group them by using the author_name because we need to select an author by name. We want the most books sold which were written by this author so we use the descending order to get the most sold books at the top. After that, as there is no limitation I will get all the columns for all languages to solve that I use the limit option so that I can obtain the number of columns I want. We wanted an author who is more reputed and got the most selling books so we decided to get the most sold out books by a single author. We can gain more popularity by choosing a famous author so that more people will pay attention to us.

Result table:

author_name	total_books
Stephen King	65

Explanation for output table:

The output table contains two columns which we have selected in the statement and we have arranged them by using the author name so it counts the total books each author has written by authors using name criteria. As we have given a limit of 1 there is only one output which we wanted.

Recommendation:

I think picking up the most famous person will increase the popularity of the company because the books sold out more were written by this author so more people are likely to follow him and admire him. So when we use him I think we can get more popularity.

3. What 10 customers should the sales team contact?

Query used

SQL statement:

```
SELECT customer.customer_id,first_name,last_name, COUNT(order_line.book_id) AS  
total_orders  
  
FROM customer  
  
JOIN cust_order ON customer.customer_id = cust_order.customer_id  
  
JOIN order_line ON cust_order.order_id = order_line.order_id  
  
GROUP BY customer.customer_id, customer.first_name, customer.last_name  
  
ORDER BY total_orders DESC  
  
LIMIT 10;
```

Explanation:

In this statement we have to select 10 customers I wanted to see the customer ID, name and the most books they have ordered so I wrote them in the select statement. I have seen that there is no table having all these data in it so I joined two tables which are customer and cust_order. We have grouped the tables with customer ID and customer first name. We have

arranged the total orders in descending order so that we can get the customer who ordered the highest books first.

Why this query?

This query takes out the top 10 persons who bought the most books from the company.

Which means these are the most valuable persons to the company. These are the customers we are getting more returns for our company.

Result table:

customer_id	first_name	last_name	total_orders
8	Nolly	Bonicelli	36
171	Lucie	Bartlam	35
87	Tricia	Saul	28
116	Shae	Leser	28
75	Enrichetta	Morrill	28
19	Sanford	Gillbe	27
16	Debbi	Huyghe	27
49	Lynda	Roseborough	24
99	Lianne	Gorry	21
131	Olga	Allcott	21

Explanation for output table:

This is the result table for the query we have written it shows the top 10 people who had ordered the most number of books from our company. There are four columns in the table which we have written in the select statement which are id, customer first name, customer last name

and the total orders placed by each customer. There are 10 rows in the table as we wanted. We have set the limit of rows to 10 that's the reason there are 10 rows in the table.

Recommendation:

The company has to reach out to these customers, provide some loyalty rewards, and ask for suggestions to improve our company. By using their personalized suggestions, we can improve our company and give them some exclusive discounts.

4. Gravity Books is looking to add a warehouse to help optimize shipping. Where should they locate that warehouse? Name the specific city and country for the location.

Query used**SQL statement:**

```
SELECT address.city, country.country_name, COUNT(cust_order.order_id) AS total_orders
FROM cust_order
JOIN customer_address ON cust_order.dest_address_id = customer_address.address_id
JOIN address ON customer_address.address_id = address.address_id
JOIN country ON address.country_id = country.country_id
GROUP BY address.city, country.country_name
ORDER BY total_orders DESC
```

LIMIT 5;

Explanation:

We have selected city, country name and total orders as there is no table with all these columns in it we are joining tables like cust_order, customer_address, address and country by the common columns. We want to group the total orders for each city and country so we have written city and country names in group by function. We want them in the descending order of total orders and we have set a limit to 5.

Why this query?

This query gives us the most number of orders placed from each city, which means there is the most demand in these areas. So, if we have to increase our supply in these areas to sell books and meet the demand, we cannot use sales as the price for each book varies.

Result table:

city	country_name	total_orders
Bauchi	Nigeria	290
Quinta	Portugal	246
Shaghat	Armenia	232
Langchuan	China	231
Jardín	Colombia	225

Explanation for output table:

This is the result table for the query we have written. We have 3 columns which we selected in

the query. We can see the total orders placed for each city with the county name which we have written in the group by condition. There are only 5 columns in this table because we have set the limit to 5.

Recommendation:

I think the company has to set up more warehouses in these locations to reach the public demand. As the number of books sold in these places is high we need to add were houses to these places. If there is a situation like there is not enough stock to deliver the books the customers might jump to another company. So to stop that we need to set a were house. By setting up warehouses in these areas we can also decrease the shipping cost and improve delivery efficiency.

Additional insights:**1. Margin analysis**

Observation: There is no profit column in the order line table.

Insight: adding an extra column of the profit table which gives us the amount of profit we got for a specific order. By adding this column we can easily calculate the margins for each book or the order so when we want to delete a book from the list we can calculate the total books sold and also the profit we got by selling that book with this we can the best way to select a book.

2. shipping cost analysis

Observation: Adding a shipping cost in the shipping method.

Insight: Adding a shipping cost analysis we can evaluate the costs for every different shipping method. By this analysis, we can find out the most cost-effective shipping options by we can further negotiate for better rates and we can analyse the most used shipping methods. The company can cut the cost of shipping and optimize the shipping methods.

3. Adding a discount for customers table in the customer table.

Observation: adding a discount table which gives discounts to special customers

Insight: when we add a discount table for customers we can analyse how many people are using the discount. This will help us to understand the impact of discounts on sales. We can also

calculate how much profit we get after the discounts. With this information, we can decide whether to continue or not to get better profits.